Aptitude

(Amit Rohara)

→ 19 → 10 + 9 → divide in nearest round figure & (add or sub) the next

6	60	1	61	59
12	120	2	122	118
18	180	3	183	177
24	240	4	244	236
30	300	5	305	295
36	360	6	366	354
42	420	7	427	413
48	480	8	488	472
54	540	9	549	531
60	600	10	610	590

I should know tables from 1 → 20

Assignment 37, 73, 92 ; 48

		37			73			92	48
30	7	37	70	3	73	90	2	92	48
60	14	74	140	6	146	180	4	184	96
90	21	111	210	9	219	270	6	276	144
120	28	148	280	12	292	360	8	368	192
150	35	185	350	15	365	450	10	460	240
180	42	222	420	18	438	540	12	552	288
210	49	259	490	21	511	630	14	644	336
240	56	296	560	24	584	720	16	736	384
270	63	333	630	27	657	810	18	828	432
300	70	370	700	30	730	900	20	920	480

$72 \times 25 \rightarrow \cancel{72}^{18} \times \frac{100}{\cancel{4}} = 1800$

__/__/__

* $68 \times 50 \rightarrow \frac{\overset{34}{\cancel{68}} \times 100}{\cancel{2}} = 3400$

* Squares

12 → 144	21 → 441	46 → 2116
13 → 169	22 → 484	
14 → 196	23 → 529	
15 → 225	24 → 576	
16 → 256	25 → 625	
17 → 289	26 → 676	
18 → 324	27 → 729	
19 → 361	28 → 784	
20 → 400	29 → 841	

$46 \rightarrow (40+6)^2 \rightarrow (40)^2 + (6)^2 + 2 \times 40 \times 6 = 1600 + 36 + 480$

$= 2116$

→ Base 50

Step 1 → get the difference & Square it

Step 2 → Subtract the same difference from 25.

For ex:- $46 \rightarrow (50 - 4)$ ↳ difference.

Step 1) → 4 → 16

Step 2) → 25 − 4 → 21

2116

$45 \rightarrow (50 - 5)$

① $5^2 \rightarrow 25$

② $25 - 5 = 20$

2025

* 44 → (50 − 6)

① 6^2 → 36

② 25 − 6 → 19

$\boxed{1936}$

* 43 → (50 − 7)

① 7^2 → 49

① 25 − 7 ⇒ 18

$\boxed{1849}$

* 47 → (50 − 3)

① 3 → 09

② 25 − 3 → 22

$\boxed{2209}$

* 48 → (50 − 2)

① 2 → 04

② 25 − 2 → 23

$\boxed{2304}$

* 49 → (50 − 1)

① 1 → 01

② 25 − 1 = 24

$\boxed{2401}$

* 39 → (50 − 11)

1 → carry

① 11 → 21

② 25 − 11 → 14 + 1 = 15

$\boxed{1521}$

* 38 → (50 − 12)

1 → carry

① 12 → 44

② 25 −12 → 13 + 1 = 14

$\boxed{1444}$

* 37 → 1369

* 36 → 1296

* 54 → (50 + 4)

① 4 → 16

② 25 + 4 → 29

$\boxed{2916}$

* 61 → (50 + 11)

①

① 11 → 121

② 25 + 11 → 36 + 1 = 37

$\boxed{3721}$

* 62 → (50 + 12)

①

① 12 → 144

② 25 + 12 → 37 + 1 → 38

$\boxed{3844}$

* 63 → (50 + 13)

①

① 13 → 69

② 25 + 13 → 38 + 1 = 39

$\boxed{3969}$

* 64 → (50 + 14)

①

① 14 → 196

② 25 + 14 → 39 + 1 → 40

$\boxed{4096}$

* Base 100

Step ① Get the difference & Square it

Step ② Add the difference to the same ~~num~~ number.

* 105 difference

6

① 105 → (100 + 5)

① 5^2 → 25

② 105 + 5 = 110

$\boxed{11025}$

* 106

106 → (100 + 6)

① 6^2 → 36

② 106 + 6 = 112

$\boxed{11236}$

* $111 \to (100 + 11)$

① $11^2 \to$ ① 21 → carry

② $111 + 11 = 122 + 1 = 123$

$\boxed{12321}$

* $112 \to (100 + 12)$

$\boxed{12544}$

* $113 \to (100 + 13)$

$\boxed{12769}$

* $96 \to (100 - 4)$

① $4^2 \to 16$

② $96 - 4 = 92$

9216

$91 \to 8281$

$92 \to 8464$

$93 \to 8649$

$94 \to 8836$

$95 \to 9025$

* $89 \to (100 - 11)$

① → carry

① $11 \to 121 \to 21$

② $89 - 11 = 78 + 1 = 79$

$\boxed{7921}$

* $88 \to (100 - 12)$

①

① $12 \to$ ① 44

② $88 - 12 = 76 + 1 = 77$

$\boxed{7744}$

* 7744 is the only perfect square, which is of type XXYY.

Q.) How many 4-digit perfect squares have the 1st two digits same and last two-digits same.

Ans = 1 → (7744)

* 87 → (100 - 13)

carry

1. 13 → 169
2. 87 - 13 = 74 + 1 = 75

$\boxed{7569}$

* 204 → ~~20816~~ (200 + 4) → 41616

* <u>Base 200</u>

1. Get the diff & Square it
2. Add the diff to the same number
3. And the multiply by <u>2</u> (i.e. the most significant bit (xxxxx))

* 204 → (200 + 4)

4 → 16

204 + 4 = 208 x 2 = 416

$\boxed{41616}$

* 207 → (200 + 7)

7 → 49

207 + 7 → 214 x 2 = 428

$\boxed{42849}$

* 209 → (200 + 9)

9 → 81

209 + 9 → 218 → 436

$\boxed{43681}$

* 211 → (200 + 11)

11 → 121

211 + 11 = 222 x 2 = 444 + 1

= 445

$\boxed{44521}$

* 2412 → (4200 + 12)

$\boxed{44944 \rightarrow xxyxx}$ unique no-

* 213 → (200 + 13)

13 → 1 69

213 + 13 = 226 = 452 + 1 = $\boxed{45369}$

* 196 → (200 − 4)

4 → 16

~~200~~ 196 − 4 = 192 × 2 = 384

$\boxed{38416}$

* 195 → (200 − 5)

5 → 25

195 − 5 = 190 × 2 = 380

$\boxed{38025}$

* 194 → (200 − 6)

6 → 36

194 − 6 = 188 × 2 = 376

$\boxed{37636}$

* 306 → (300 + 6)

6 → 36

306 + 6 = 312 × 3 = 936

$\boxed{93636}$

* 308 → (300 + 8)

8^2 → 64

308 + 8 = 316 × 3

= 948

$\boxed{94864}$

* 312 → (300 + 12)

12 → 1 44

312 + 12 = 324 × 3

= 972 + 1 = 973

$\boxed{97344}$

200 - 8

64

__/__/__

$\begin{array}{r} 192 \\ -8 \\ \hline 184 \end{array} \times 2 = 368$

* 409 ⟶ (400 + 9)

9 ⟶ 81

409 + 9 ⟶ 418 × 4

= 1672

167281

* 34 = 1156

67 = 4489

81 = 6561

118 = 13924

215 = 46225

192 = 36864

315 = 99225

316 = 99856

(23716) ⟵ 154 ⟹ (150 + 4) ⟶ $\left(\frac{3}{2} \times \cancel{158}^{79}\right)(16)$ (154 + 4)

255

(23716)

$\left(\left(\frac{5}{2}\right) \times \cancel{260}^{130}\right)(25)$

$\boxed{65025}$

* Each multiple of 50 is considered as a Base.

5 ⟶ 25

15 $\xrightarrow{\times 2}$ 225

25 $\xrightarrow{\times 3}$ 625

35 $\xrightarrow{\times 4}$ 1225

45 $\xrightarrow{\times 5}$ 2025

55 $\xrightarrow{\times 6}$ 3025

* 135 $\xrightarrow{13 \times 14}$ 18225

165 $\xrightarrow{16 \times 17}$ 27225

* $\sqrt{123454321}$

1 → 1

11 → 1 2 1

111 → 1 2 3 2 1

1111 → 1 2 3 4 3 2 1

11111 → 1 2 3 4 5 4 3 2 1

The middle digit denotes the digit, no. of digits in the number.

* find square of 111111111 → 1 2 3 4 5 6 7 8 9 8 0 9 8 7 6 5 4 3 2 1

* Square Roots

2	7056
2	3528
2	1764
2	882
21	441
21	21
	1

Pick only half factors, we get the square root

→ $2 \times 2 \times 21$

$= \boxed{84}$

2	5184
2	2592
2	1296
2	648
18	324
18	18
	1

$= 2 \times 2 \times 18$

$= 72$

__/__/__

* ~~6689~~

→ The last digit of a Square is Never 2, 3, 7, 8 and it is always 0, 1, 4, 5, 6, 9.

* Which of the following numbers can be a perfect Squares?

1. 99998 ×
2. 98182 ×
3. 96363 ×
4. 98596 ✓

* Which of the following can be a perfect square?

1. 49186 ×
2. 47366 ×
3. 43326 ×
4. 4~~7~~6656 ✓

→ If the last digit of a Square is 6, then 2nd last digit is always an odd number. (1, 3, 5, 7, 9)

→ If the last digit of a Square is 5, then the 2nd last digit is always, 2.

→ If the last digit of a Square is 1, then the second last digit is always even number (0, 2, 4, 6, 8)

↳ the same fact is true for 4 & 9

_/_/__

* The no. of a zeros, at the end of a square is always even and the non-zero part should be a perfect square

* 47 | 61 → firstly ignore last 2 digits

36   49 → Secondly find the range for 47 in b/w the squares

↓   ↓

60   70 → Write the square roots of those squares, now

61/69 → Squares whose last digit ends with 1

↓

now 47 is closer to 49 so, we will go with 69

* 68 | 89

64   81

↓   ↓

80   90

83/87 → 68 is closer to 64, so we go with 83

* 6724

64   81

↓   ↓

80   90

82/88 → 82

* 3481

25   36

↓   ↓

50   60

~~51~~/59 → 59

* 7921 → 89

* 6241 → 79

* 5329 → 73

__/__/__

* 4120 9

  $\underbrace{412}$ 09 → ignore firstly

→ find range b/w squares

① 400 - 441

→ find the square roots.

② 200 - 210

→ find no's whose squares last digit ends in 9 are

→ 203/207

→ Now, 412 is closer to 400 than 441 so we go with 203.

* 23104

  225 256

  ↓ ↓

  150 160

  → 152/158

  152

* 231.04 → 15.2

* 4.1209 → 2.03

* For a square, the no. of digits after decimal point is always even.

__/__/__

* 0.7788 → 0.88

* 1.4641 → 1.21

* Find the no. of non-Square numbers b/w 202^2 and 203^2.

$1^2 (1)$ $2^2 (4)$ → 2 (non-perfect sq.)
$2^2 (4)$ $3^2 (9)$ → 4 (— " —)
$3^2 (9)$ $4^2 (16)$ → 6 (— " —)
$4^2 (16)$ $5^2 (25)$ → 8

Similarly, for the numbers of non perfect squares b/w two no's squares is twice / double of smaller number, i.e.

$1^2 - 2^2 \rightarrow 1 \times 2 = 2$
$2^2 - 3^2 \rightarrow 2 \times 2 = 4$
$3^2 - 4^2 \rightarrow 3 \times 2 = 6$
⋮ ⋮
$202^2 - 203^2 \rightarrow 202 \times 2 = \boxed{404}$

* Find the Smallest number by which 2800 should be multiplied or divided to get the result as a perfect square.

a) 2 b) 4 c) 5 d) 7

7

→ After factorisation, we get that only 7 occurs 1 time, so, to make it perfect square, we should make it either even or zero. So, the answer is 7

Que wrong

(X) * Find the smallest natural no. that must be added to 1030 to make it a perfect square.

* $\sqrt{11881} \times \sqrt{x} = 10137$
(109)

a) 8281
b) 8649 ✓
c) 9216
d) 9409
e) NOTA

* $\sqrt{36 + 18\sqrt{3}}$

a) $2 + 5\sqrt{3}$
b) $3 + 3\sqrt{3}$ ✓
c) $5 + \sqrt{3}$
d) more
e) none

→ make the internal value a perfect square, i.e.

$\sqrt{(3+3\sqrt{3})^{\cancel{2}}} = 3 + 3\sqrt{3}$

* Find the smallest no. that must be added to 6072 to make it a perfect square?

① 6 × → adding 6 makes last digit 8
② 10 × → — " — 10 ———— " ———— 2
③ 12 ✓
④ 16 × → — " — 16 ———— " ———— 8

* Find the smallest natural no which must be added to 4523, so that the sum is a perfect square

(a) 97 → 4620 → cant have single last digit 0.
(b) 99 → last digit becomes 2
(c) 101 → ✓
(d) 103 → 4626 → ending with 6, must have last 2^{nd} digit as (1, 3, 5, 7, 9)

* Find the square root of $\sqrt{1471369}$

(1) 1203
(2) 1233
(3) 1213
(4) 1223

13 has a square of 169, so, we'll choose the number which contains 13.

* If the last 2 digits of number is 69, then the square ~~number~~ root of number is of type $50x \pm 13$.

* $\sqrt{211.9936}$

(1) 14.86
(2) 15.44 ×
(3) 14.56 ✓
(4) 15.84 ×

211
196 225
14 15 → so answer will be (14 - 15)

$\sqrt{36}$
6 so, possible options 06, 44, 56, 94

so, $\boxed{18.56}$ is the answer.

_/_/_

* $\sqrt{7084}$

① 78 ② 76 ③ 82 ④ 64

60

49 64

↓ ↓

70 80

72/78

* Square root of 291600

291600 → 29|16 $\times (10)^2$

25 36

↓ ↓

50 60

↓

54/56

$(54 \times 10)^2$

$\sqrt{291600} = 540$

a) 520

b) 620

c) 580

d) 540

* evaluate $\sqrt{0.9}$ up to 3 places of decimal.

$0.9000 \rightarrow 9000 \times 10^{-4}$

$(0.95)^2 \times 10^{-3} \rightarrow 9025 \times 10^{-4}$

the no.'s are closer, so our answer will be, < 0.95 and > 0.94

a) 0.948 b) 0.958 c) 0.938 d) 0.978

* 8160 Soilders $\longrightarrow$ forms a Square formation,
But 60 soilders are left out, so, find the sq. side.

$8160 - 60 = 8100$

a) 96 b) 92 c) 90 d) 99

* Find the smallest no. by which 28812 must be divided so that the quotient becomes a perfect sq.

a) 9 b) 7 c) 6 d) 3

$28812 = 2 \times 2 \times 3 \times 7 \times 7 \times 7 \times 7$

odd one out.

* Sq. root of $0.289 / 0.00121$

$(289/121) \times 10^{1}$

$(170/11)^2 \times 10^{1}$

$\frac{17}{11} \cancel{\times 10} = \left(\frac{170}{11}\right)$

* least sq. no. which is exactly divisible by each one of no. 15, 18, 12

a) 1800 b) 3600 c) 2400 d) 4200

only sq. no. available.

* Sqrt (256.6404)

$\downarrow$ 16 $\downarrow$ 2/8

a) 16.02 m b) 17.02 m c) 18.02 m d) 19.02 m

* least no. to be added to 7900, to make it a perfect square

a) 67 b) 87 c) 97 d) None of above

(cause no perfect sq. ends in 7)

* Smallest 3 digit Perfect Square :- 100
* Greatest 3 digit Perfect Square :- 961 (31^2)
* Smallest 4 digit Perfect Square :- 1024 (32^2)
* Greatest 4 digit Perfect Square :- 9801 (99^2)
* Smallest 6 digit Perfect Square :- 100489 $(317)^2$
* Greatest 5 digit —— " —— :- 99856 $(316)^2$
* Smallest —— " —— :- 10000 $(100)^2$

* Puzzles

Que) 3 choco → ₹1

for each 3 wrappers 1 choco

Return max. no. of choco. that can be purchased, provided no wrapper are left.

$10 \times 3 = 30 + 30/3 = 10$

$= 30 + 10/3$

$= 30 + 10 + 3/3$ (1)

$= 30 + 10 + 3 +$ \$1 (2)

$= 44$ (2)

$= 44 + 1 = 45$ → wrappers

(Now, we take a chocolate from shopkeeper now, we have 3 wrappers and pay the shop for taken choco)

Que) You have to send a Parcel from Point A → B without Sending the Key to Parcel by the hands of Postman. How do you send it?

Cryptography (Public & Private key) or

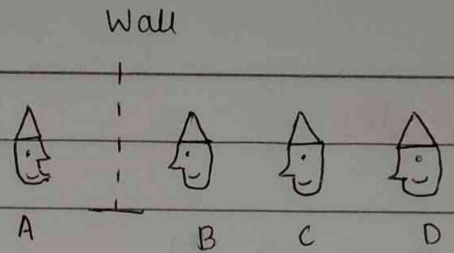
A locks and send to B, then (A lock)

B locks and send back to A, then (A + B lock)

A unlocks with his lock & sends to B, then (B lock)

B can now unlock using his key as only B's lock is present.

__/__/__



* 4 Hats

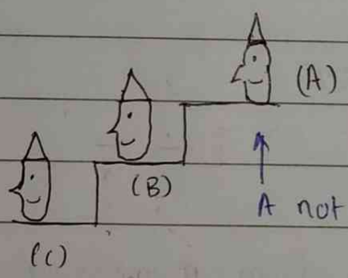
2 white   2 Black

→ in 1st 1-minute, D doesn't speak so, which means B & C are wearing diff. color hats.

in 2nd minute, C needs to speak, about the color.

Ans. is C should Speak.

* Or,



A not speak, so following

5 Hats

3 B   2 W

B	B
B	W
~~W~~	~~B~~
~~W~~	~~W~~

Numbers

* $N = \{1, 2, 3, \ldots\}$ $x+1 = 1$ (Natural)

* $W = \{0, 1, 2, 3, \ldots\}$ $x+2 = 1$ (Whole)

* $I = \{\ldots, -3, -2, -1, 0, 1, 2, 3, \ldots\}$ $2x = 1$ (Integers)

* Rat, $Q = \{ p/q \mid p, q \in I, q \neq 0 \}$ (Rational)

* Irrational Numbers $\rightarrow$ not Rational (Irrational)
  $(\bar{Q}) \rightarrow \sqrt{2}, \sqrt{3}, \pi, e, \ldots$

* $R = Q \cup \bar{Q}$ $\longrightarrow$ Real Numbers

→ 0 is an even no. as it is divisible by 2.

→ A no. having only/exactly two different factors is a Prime number.

* 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53,
  59, 61, 73, 79, 83, 89, 97
  ,67,

* There are 25 prime numbers from 1 - 100.

* A no. having more than 2 factors is composite number.

* 1 is neither prime nor composite.

* 2 is the only even-prime number.

__/__/__

* Except (2 & 3), every prime no. is of the type,

$\boxed{6x-1 \,/\, 6x+1}$

* find all the Prime no.'s in b/w 100 & 200. (21)

101, 103, 107, ,109, 113, 127, 131, 137, 139, 149, 151, 157, 163, 167
173, 179, 181, 191, 193, 197, 199.

* 1296 → 36

→ There are 16 Prime numbers b/w 200 & 300.

* Total no. of students in school is prime no. which of the following can be the ratio of girls : boys in school?

a) 86 : 81
b) 70 : 73
c) 80 : 81
d) 90 : 91

(2 3) (5 7) (11 13) (17 19) (29 31) (41 43) (59, 61) (71, 73) ......

* 4, 6, 12, 18, 30, 42, 60, 72, 102, 108, ... (Twin Primes)
2 & 3 6 9 15 21 30 36 51 54

Next 2 elements are 138 & 150

Twin primes are prime no.'s whose difference is 2. they are (consecutive prime & consecutive odd)

__/__/____

* There are 3 mislabeled jars, with apples & oranges in the first and second respectively. The third jar contains a mixture of apples & oranges. You can pick as many ~~as~~ fruits as required to precisely label each jar. Determine the minimum no. of fruits to be picked up in the process of labelling the jars.

→

$\boxed{A}$ $\boxed{O}$ $\boxed{M.}$

Pick one from M if it is a apple put ~~it~~ ~~is~~ $\boxed{A}$ label on $\boxed{M}$ box

* Divisiblety ~~by~~ test

4 → last 2 digits must be divisible by 4.
5 → the ~~at~~ ~~the~~ last digit should be 0, or 5.
6 → should be divisible by 2 & 3 both.
8 → last 3 digits must be divisible by 8.
9 → Sum of digits divisible by 9.
10 → last digit must be 0.
11 → diff b/w 2 groups of alternate digits should be either 0 or a multiple of 11.

$\bar{8}1\bar{8}4$

$8+8=16$ $4+1=5$

$16-5=11$

↳ divisible by 11.

12 → the no. should be divisible by 3 & 4.
15 → ———— " ———— 3 & 5

7 → the diff b/w ~~the~~ two alternate groups taking 3 digits at a time should be either 0 or a multiple of 7.

* 7370356

356 + 7 = 363

370 − 363 = 7 → divisible.

* 23100

100 − 23 = 77 → divisible

* 550500006

550 + 006 = 556

556 − 500 = 56 → divisible by 7.

* a number abc is divisible by 7, if ab + 5c is divisible by 7.

* 14 → the no. should be divisible by 2 & 7.

* 13 → the same test that was used for 7, but here the final diff should be divisible by 13.

* abc is divisible by 13, if (ab + 4c) is divisible by 13

504 → 50 + 4(4) = 516 → ✗

✗

_/_/_

* 17 → abc is divisible by 17, if (ab - 5c) is divisible by 17.

544 → 54 - 4×5 = 54 - 20
= 34 divisible by 17.

* 19 → abc is divisible by 19, if (ab + 2c) is divisible by 19.

551 → 55 + 2×1
→ 57 → divisible by 19

551 ✓

* 720720 → is the smallest natural no. divisibly by the first 16 natural numbers.
(1 - 16)

* 7 digit no. 765432x / 18 , find x ?

a) 0 ✓ b) 3 c) 6 d) 9

* 6 digit no. 12345x / 12 find x ?

a) 0 b) 3 c) 6 ✓ d) 9

* find the least perfect square that is divisible by 21, 36, 66
(21 → 3, 7; 36 → 9, 4; 66 → 11, 6)

a) 213444 ✓
b) 214344 ✗
c) 214434 ✗
d) 231444 ✗

① apply divisiblity test of 11 (2 options get elim)
② apply —— " —— 4 (c gets elim)

* 10 logical Prisoners, in King's Jail.

10 hats → random Black / White Hats.

(0, 10) or (1, 9) or (2, 8) or ... (9, 1) or (10, 0)

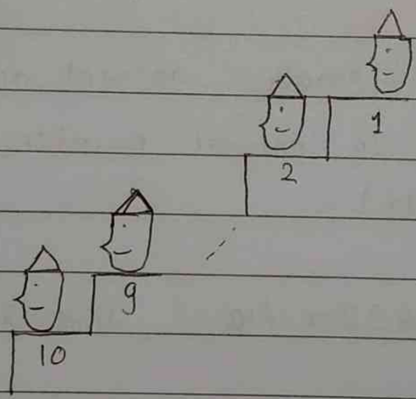
Condⁿ to be free is → All guesses by prisoners of there hat is correct (10 correct)

or

(9 correct & 1 Wrong)

If 1 Sees, an even number of Black Hats he shouts "Black"

If 1 Sees, an odd ——— " ——— he shouts "White"



* 5 Important formulas

① $1+2+3+\dots+n \rightarrow \boxed{\dfrac{n(n+1)}{2}}$

② $1^2+2^2+3^2+\dots+n^2 \rightarrow \boxed{\dfrac{n(n+1)(2n+1)}{6}}$

③ $1^3+2^3+3^3+\dots+n^3 \rightarrow \boxed{\left[\dfrac{n(n+1)}{2}\right]^2}$

④ $1+3+5+7+9+\dots+25$

* Sum of first n odd numbers = n^2

$$n = \frac{\overset{\text{last number}}{l.n.} + 1}{2}$$

5) $2 + 4 + 6 + 8 + \ldots\ldots$

Sum of first n even numbers = $n(n+1)$

$$n = \frac{l.n}{2} \rightarrow \text{last number}$$

* $21 + 22 + 23 + \ldots + 40 \rightarrow 610$

* $11 + 13 + 15 + \ldots + 29 \rightarrow 200$

* $2^2 + 4^2 + 6^2 + \ldots + 20^2 \rightarrow 2^2(1^2 + 2^2 + \ldots + 10^2) \rightarrow 1540$

* In a party of 25 people, each person shakes hands with all the other ~~people~~ people exactly once. Find total no. of hand shakes

→ No. of handshakes in a party of n people = $\frac{n \times (n-1)}{2} = {}^nC_2$

* In ICC world cup 2023, 10 teams have participated, each team play against opponent team exactly once. How many basic matches were played?

$$\frac{n \times n-1}{2} = \frac{10 \times (9)}{2} = 5 \times 9 = 45$$

__/__/____

* In a Knockout tennis tournament of 64 players, find the total no. of matches req^d to decide the winner.

$\Rightarrow 32 + 16 + 8 + 4 + 2 + 1$

$\Rightarrow 63$

For n-players, n-1 knockout matches will be played.

* Count the no. of squares on a chessboard?

$$\text{no. of squares} = \frac{n(n+1)(2n+1)}{6} \quad \text{or Sum of squares}$$

$$= \frac{\overset{4}{\cancel{8}} \times \overset{3}{\cancel{9}} \times 17}{\cancel{6}\cancel{2}}$$

$$= 12 \times 17 = \underline{204}$$

* Number of Rectangles $= \left[\frac{n(n+1)}{2}\right]^2 = \left(\frac{8(9)}{2}\right)^2 = (36)^2$

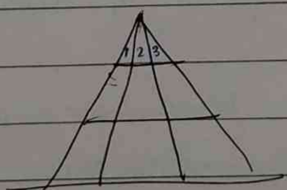
$= 1296$

* Number of Triangles in figure?

6

6

6



$3 + 2 + 1 = 6$

$\underline{6 \times 3} = 18$

__/__/__

* Salman & Jackie marry each other and have x children of their own. Salman already has y children from his 1st Wife. Jackie also has z children from her 1st husband.

$x+y+z=9$, $(x \neq y \neq z)$, find the max. number of fights that can take place among the children.

Rule:- ① If for a fight we need exactly 2 children and no two children of the same parent fights.

→ to maximize,

$$x \times y + y \times z + z \times x = \text{ans.}$$

$$x + y + z = 9$$
$$(2) \quad (3) \quad (4)$$

Or total fights - (fights of same parents)

$$^9C_2 - {}^2C_2 - {}^3C_2 - {}^4C_2$$

$$9 \times 4 - 1 - 3 - 6$$

Ans → $36 - 10 = 26$

__/__/__

* 100 prisoners present in King's Jail.

There are hats wore by each prisoner, either Black / Red.
(Not Equal (50-50))

Cond^n :- Prisoners assemble in a ground having some light.
Prisoners can't take anywhere if done no freedom.
No gestures allowed.
Hat has been glued to head for 1 day.

If Prisoners segregate them in two groups of Red / Black they can be free.

→ the condition should be,

① Prisoners come out one-by-one
② If the prisoner sees a group of same colour, the prisoner goes and stand in either right or left.
OR
③ If the prisoner sees a seperation i.e. Black Red, then the prisoner goes and becomes the new border so it doesn't disturb the groups.

_/_/_

* HCF & LCM

HCF (56, 84) = 28

HCF (72, 120, 168) = 24

HCF (259, 407)

(259, 407-259) = (259, 148)

= (111, 148)

= (111, 37)

(74, 37)

* Find HCF (72, 125) = 1

* If nothing is in common then the HCF is 1 & no. are said to be co-primes.

* Prime numbers are always co-prime.

(3,5) → 1 HCF (5,7) $\xrightarrow{\text{HCF}}$ 1

* Consecutive Natural no.'s are always co-prime

* Consecutive odd numbers are always co-prime.

* HCF of consecutive even numbers is always 2.

_/_/_

* HCF & LCM

HCF (56, 84) = 28

HCF (72, 120, 168) = 24

HCF (259, 407)

(259, 407−259) = (259, 148)

= (111, 148)

= (111, 37)

(74, 37)

* Find HCF (72, 125) = 1

* If nothing is in common then the HCF is 1 & no. are said to be co-primes.

* Prime numbers are always co-prime.

(3,5) → 1 HCF (5,7) → 1 HCF

* Consecutive Natural no.'s are always co-prime

* Consecutive odd numbers are always co-prime.

* HCF of consecutive even numbers is always 2.

* Find the biggest number such that both 42 & 98 are both divisible by that no.

HCF = 14

* Find the largest number such that if 43, 65, and 109 are divided by that number then we get the remainders as 1, 2 & 4.

HCF (42, 63, 105) = 21

In an HCF problem, firstly adjust, & then find the HCF.

* Find the greatest number such that 59, 95, 185 are divided by the number, then get the same remainder.

59 95 185

36 90

Now, find HCF of (36, 90) = 18 is the answer.

* Find the HCF of 111.....(35t)
1111.....(60t)

HCF (111...35t, 1111...60t) = 11111

		HCF
1	11 →	1
11	111 →	1
11	1111 →	11
11	1111 →	11
11	111111 →	11

Answer is hcf of no. of times 1 occur in 1st & 2nd number.
So, HCF (35, 60) = 5

_/_/_

* HCF (333 .... (60 t),
333 .... (100 t)) = 333 .... (20 t)

* HCF ($3^{45} - 1$, $3^{120} - 1$) = $3^{15} - 1$

HCF (45 , 120) = 15

* Find the HCF of ($7^{20} - 1$, $7^{20} + 1$) $\quad 7^{20} \Rightarrow$ is odd

$7^{20}-1 \downarrow$ even no. $\quad 7^{20}+1 \downarrow$ even no. $\quad$ even odd even

So, HCF of 2 consecutive even numbers is 2.

* Find HCF of ($8^{20} - 1$, $8^{20} + 1$)

$8^{20} \rightarrow$ even no.

even − 1 → odd
even + 1 → odd.

∴ 2 consecutive odd no. HCF is 1

HCF ($8^{20} - 1$, $8^{20} + 1$) = 1

* The total no. of admissions for mech., CSE, ECE branches were 48, 84 & 72 respectively, Principal wants to arrange this students into separate classrooms, so that each class contains an equal no. of students. Find the min. no. of rooms required.

HCF (72, 84, 48) = 12 $\quad$ 48 ↓ $\quad$ 84 ↓ $\quad$ 72 ↓

Imp

* How many pairs of +ve integers (1, 2, 3, ---) x & y exists so tha $x+y = 156$ and HCF of $x, y = 13$

$$13a + 13b = 156 \quad \& \quad HCF(a, b) = 1$$

$$a+b = 12$$

$\downarrow$

Now find pairs whose sum is 12.
co-prime

So, pairs is $(1, 11)$ & $(5, 7)$

* How many pairs of +ve integers x & y exist. So that $x+y = 80$ and $HCF(x, y) = 5$

$$5x + 5y = 80 \qquad HCF(x, y) = 1$$

$$x + y = 16$$

4 pairs = $(5, 11)$ $(13, 3)$ $(9, 7)$ $(1, 15)$

* Product of 2 numbers is 4107, & their HCF is 37, find the greater number.

$HCF(a, b) = 1$, $x = 37a$, $y = 37b$

$$\cancel{37}a \times \cancel{37}b = \cancel{4107}^{\cancel{111}\ 3}$$

$$a \times b = 3$$

$a = 1$ & $b = 3$ $\rightarrow$ $x = 37$, $y = 111$

_/_/_

* LCM (least common multiple)

HCF × LCM = Product of 2 nums

* LCM(36, 48) = 144

HCF ≤ x, y ≤ LCM

* LCM(20, 24, 30) = 120

* LCM(259, 407) = $\frac{\cancel{259}^{7} \times 407}{\cancel{37}}$

= 407 × 7

= 2849

* LCM(203, 309) = $\frac{203 \times \cancel{309}^{11}}{\cancel{29}}$ = 2233

* For co-prime numbers, the product is the LCM.

* LCM(20, 24, 30) = 120

* LCM(15, 36, 45) = 180

* Find the least no. which when divided by 20, 16, 12 leaves a common remainder of 7 in each case.

In an LCM problem, first find the LCM & then adjust.

LCM(20, 16, 12) = 240

Now add 7 to ~~[illegible]~~ LCM → 7 + 240 = 247

* Find the smallest no. which when divided by 24, 36, 45, leaves remainders as 17, 29, 38 respectively.

LCM (24, 36, 45) = 360

LCM - common difference

24 - 17 = 7
36 - 29 = 7 → common diff.
45 - 36 = 7

360 - 7 = 353

* Find smallest no. which when divided by 4, 5, 7 leaves remainders as 2, 3, 1 respectively.

a) 392 b) 218 c) 358 d) NOT

* Whenever we add or subtract the LCM, then the basic properties of a no. do not change.

* CRT → chinese remainder theorem

4	5	7
2	3	1

7+1 = 8

5+3 = 8

Now take lcm (5,7) = 35

LCM

Now, add to 8 until the condition for 4 gets satisfied.

$35 + 8 = 43 \ \% \ 4 \neq 2$

$43 + \overset{35}{\cancel{8}} = 78 \ \% \ 4 = 2$ ✓ Satisfied

* Find smallest no. which when divided by 7, 8, 10 leaves remainder as 1, 4, 2

$10 + 2 = 12$

$8 + 4 = 12$

$LCM(10, 8) = 40$

~~52~~ $52 \ \% \ 7 \neq 1$

$52 + 40 = 92 \ \% \ 7 = 1$

$\therefore$ answer = 92

* NO. % 5, 6, 7 $\longrightarrow$ 3, 4, 2

5	6	7
3	4	2

$7 + 2 = 9 + 7 = 16$

Now, $16 \ \% \ 6 = 4$ $\rightarrow$ Satisfies

Now $LCM(6, 7) = 42$

$16 + 42 = 58 \ \% \ 5 = 3$

Answer = 58

_/_/_

* Least no. when divided by 5, 6, 7, 8 → remainder 3 as common remainder, but when divided by 9 leaves no remainder.

a) 1677 ✓b) 1683 c) 2523 d) 3363

↓

Sum of no. is divisible by 9.

* Find Smallest no. which when divided by 16, 18, 20, 25, leaves 4 as a common remainder, but when divided by 7 leaves no. remainder.

a) 17004 b) 18000 c) 18002 ✓d) 18004

$17-4 = 13$ ✗ $\% 7 \ne 0$ $18-0=18 \% 7 \ne 0$ $18-2=16 \% 7 \ne 0$ $18-4=14 \% 7 = 0$

* Find Smallest multiple of 23, which when divided by 18, 21, 24 leaves remainders as 7, 10, 13 & respectively.

a) 3002 ✓b) 3013 c) 3024 d) 3036

Divisor → Even

Remainder → Odd

then, the original number is also odd.

__/__/__

* 3 bulbs, Red, ~~Green~~ Orange, green, flashes after 20, 30, 40 seconds then after how much seconds they will blink at same time.

LCM (20, 30, 40) = 120 seconds.

→ In 30 minutes how many times will they blink at same time.

30/2 = 15 times

* Find the HCF & LCM of $\left(\frac{12}{25} \ \& \ \frac{16}{15}\right)$

$$\boxed{\text{HCF of fractions} = \frac{\text{HCF of Numerators}}{\text{LCM of Denominators}}}$$

~~LCM of fractions = LCM of Numerators / HCF of Denominators~~

~~$lcm\left(\frac{12}{25}, \frac{16}{15}\right) = \frac{48}{8}$~~

$$\boxed{\text{LCM of fractions} = \frac{\text{LCM of Num}}{\text{HCF of Denom}}}$$

find the Greatest

$2^{1/2}$, $\boxed{3^{1/3}}$, $4^{1/4}$, $6^{1/6}$

multiply powers with lcm (2, 3, 4, 6) = 12

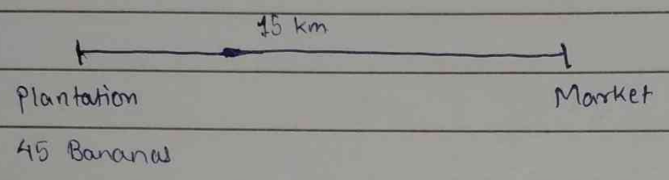
2^6 , 3^4 , 4^3 , 6^2

64 81 64 36

Aptitude Continued

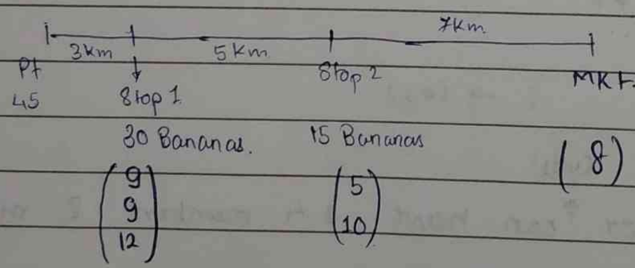
* Puzzle

- A banana plantation ~~o~~ owner has 45 plantations & the market is 15 km far, he has a Camel which can only goes 15 km one at a time and for each km covered camel eats a Banana provided camel doesn't come back. Find the maximum no. of bananas camel can reach with the end.



Camel → 1 km → 1 B

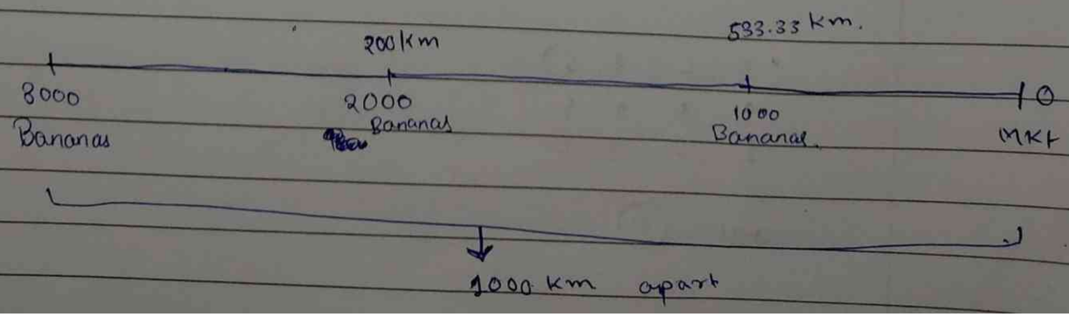
→ we'll make a stopover of 3 km as 1^{st} stop, which makes our 1 stop less.



(8) Bananas (optimal Solution)

1^{st} Stop : $\frac{45 B}{15} = 3$ km

2^{nd} Stop :



Power Cycles

→ 2 → (2, 4, 8, 6) → last digit 's follow a pattern called as Power cycle.

→ 3 → (3, 9, 7, 1)

→ 4 → (4, 6)

→ 5 → (5)

→ 6 → (6)

→ 7 → (7, 9, 3, 1)

→ 8 → (8, 4, 2, 6)

→ 9 → (9, 1)

→ 0 → (0) 1 → (1)

At max a power cycle can have 4 members & min (1) member

→ Find the last digit of 2^{17}. (2)

$17 \% 4 = 1$

4 = power cycle size

(2, 4, 8, 6)

1 2 3 4

ans = 2.

→ 2^{43}

$(43 \% 4) = 3$ $(2\ 4\ 8\ 6)$

→ 2^{100}

$(100 \% 4) = 0$

$$\begin{array}{cccc} 1 & 2 & 3 & 0 \\ 2 & 4 & 8 & 6 \end{array}$$

→ 3^{41}

$(41 \% 4) = 1$

$$\begin{array}{cccc} 1 & 2 & 3 & 0 \\ 3, & 9, & 7, & 1 \end{array}$$

→ 3^{23}

$(23 \% 4) = 3$ $(3\ 9\ 7\ 1)$

→ 3^{39}

$(39 \% 4) = 3$ $(3\ 9\ 7\ 1)$

→ 7^{63}

$(63 \% 4) = 3$ $(7\ 9\ 3\ 1)$

→ 8^{24}

$(24 \% 4) = 0$

$$\begin{array}{cccc} 1 & 2 & 3 & 0 \\ 8 & 4 & 2 & 6 \end{array}$$

→ 4^{22}

$(22 \% 2) = 0$

$$\begin{array}{cc} 1 & 0 \\ 4, & 6 \end{array}$$

→ 5^{25}

$(25 \% 5) = 0$ (5)

→ 9^{21}

$(21 \% 2) = 1$

$$\begin{array}{cc} 1 & 0 \\ 9, & 1 \end{array}$$

* Find last digit of $2^{21} + 3^{41}$

last digit of 2^{21} + last digit of 3^{41}

$\downarrow$ $\downarrow$

2 3

→ $2+3 = 5$

$\downarrow$

answer

* last digit of $9^9 + 5^5$

$\downarrow$

9 5

1 → carry

$9 + 5 = 4$

$\downarrow$

answer

* $9^{10} - 5^5$

$\downarrow$ $\downarrow$

1 5

$11 - 5 = 6$

* $2^{23} \times 7^{13}$

$\downarrow$ $\downarrow$

8 7

$8 \times 7 = 56$

$\downarrow$

last digit (answer)

(does not work for division as division starts from 1st digit (highest power i.e. leftmost bit))

* 7 is the only number which has a power cycle for last 2 digits i.e.

(07, 49, 43, 01)

__/__/____

→ Find remainder when 7^{27} is divided by 10.

$7^{27}/10 = 3$

$(27 \% 4) = 3$

1 2 3 0

(7 9 3 1)

→ Find remainder when 7^{27} is divided by 100.

$7^{27}/100$

a) 23

~~b)~~ 43

c) 63

d) 83

→ Find last 2 digits of 7^{2008}

<u>01</u>

2008 % 4 = 0

(7 9 3 1)

last digit

→ Find remainder 7^{22} is divided by 8.

(First 4 powers = GAME OVER)

7/8 = 7

49/8 = 1

343/8 = 7

2401/8 = 1.

Pattern

So, odd power = 7

even power = 1

Answer = 7

→ find remainder when 4^{43} is divided by 5

$4 \% 5 = 4$ odd power : 4

$16 \% 5 = 1$ even power = 1

$64 \% 5 = 4$

$256 \% 5 = 1$

→ find remainder when $4^{96} / 6$.

$4 \% 6 = 4$ remainder is always 4.

$16 \% 6 = 4$

$64 \% 6 = 4$

→ find remainder when $5^{39} / 4$

$5 \% 4 = 1$ remainder is always 1.

$25 \% 4 = 1$

$125 \% 4 = 1$

* Whenever any power of $(x+1)$ is divided by (x) then the remainder is always 1.

→ find remainder $5^{36} / 24$

$$5^{36} \rightarrow (5^2)^{18} \rightarrow \frac{(25)^{18}}{24} = 1 \text{ remainder}$$

→ find remainder when $7^{84} / 342$

$$(7^3)^{28} = \frac{343}{342} \rightarrow 1 \text{ remainder.}$$

→ find remainder when $2^{256} / 17$

$\frac{(16)^{64}}{17} \longrightarrow$ 1 answer.

* Whenever any power of x is dividided by $(x+1)$ then the remainder follows a pattern of,

remainder $\rightarrow$ x — (odd power)
1 — (even power)

* Find the remainder when $12^{36} / 19$

FLT → Fermat's Little Theorem.
Let a be natural no. & b is prime no. such that a is not divisible by b then ~~the remainder~~ when a^{b-1}/b, then remainder is always 1.

$12^{18}/19 = 1$.

So, $12^{36}/19 = 1$

* for 4 consecutive natural numbers a, b, c, d,

$(a^3 + b^3 + c^3 + d^3)$ is always divisible by $(a+b+c+d)$

* $11^{25}/7 \longrightarrow (11 \% 4)^{25}/7 \rightarrow (4)^{25}/7$.

pattern finding for 1st (4 powers)

_/_/_

* $(13^{100} + 17^{100}) / 25 \rightarrow$ remainder ?

$...1 + ...1 = (...2)$ num $/25$

When a number consisting last digit as 2 and divided by 25, then the last digit is ~~always~~ of remainder is always 2 & 7.

* $r1 + r2 - r3 \rightarrow$ answer

* <u>Puzzle</u>

13 P , 15 Y , 17 M → Sample down.

How to achieve, if all the chameleons can be of same color at end.

1 P , 3 Y , 5 M

1P \ / 1Y

2Y 7M.

$\boxed{Y+P=M}$

Solution is, if any 2 colors of chameleons are in same no. of quantity or all 3 colors of chameleons must be in same no. of quantity, then only it is possible, otherwise it's not possible

* even → 20 B, odd → 13 R

B → Black

R → Red

if 2 same Black Balls = add 1 Black

if 2 diff balls or same Red balls = add 1 Red.

At the end which Ball will remain ?

→ Sample down. → (even) 2 B → 1 B (nikale) }
(odd) 1 R → 1 R } 2 nikale → (1 R) → lastly 1 Red Ball remain

* Average

$$Av = \frac{Sum}{n}$$

* 1 ... 20

$$\frac{(20 \times 21)}{2} = \frac{210}{20} = 10.5$$

* 2, 4, 6, ... 30.

$$n = \frac{30}{2} = 15 \qquad Sum = n \times n+1$$

$$= 15 \times 16$$

$$= 240$$

$$\frac{240}{15} = 16$$

* Arithmetic Progression.

$$S_n = \frac{n}{2}\left[2a + (n-1)d\right] \qquad n = \frac{l.n - f.n}{d} + 1$$

$$= \frac{\text{last no.} - \text{first no.}}{d} + 1$$

$$d = n_2 - n_1$$

for an AP, average is $\boxed{\frac{\text{First term} + \text{last term}}{2}}$

4, 9, 14, ... , 94 → $\frac{94+4}{2} = \frac{98}{2} = 49$

* find avg of all add no's b/w 100 & 500

$$\frac{101+499}{2} = \frac{600}{2} = 300$$

* $X = \{1, 3, 5, \ldots, 2007\} \rightarrow Avg \rightarrow A = \frac{(1+2007)}{2} = 1004$

  $Y = \{2, 4, 6, \ldots, 2008\} \rightarrow Avg \rightarrow B = \frac{(2+2008)}{2} = 1005$

$B - A = ?$

$1005 - 1004 = 1$

→ Replacement

→ avg. weight of class of 30 students is 58 kg. two students weighing 47 & 51 kg were replaced by 2 new students 70 & 88 kg respectively

$$\frac{1740 - (47+51) + (70+88)}{30} = \frac{1800}{30} = 60 \text{ kg}$$

(Profit & loss is shared among all)

→ Avg. marks of 84 students is 62 marks, 2 students with 90 & 92 marks were replaced by ~~90~~ 60 & 80 respectively.

$(90 - 60) + (92 - 80)$

$30 + 12$

42 (loss)

$\frac{42}{84} = \frac{1}{2}$ (0.5)

$62 - 0.5 = 61.5$

* 10 yrs ago, total age of 8 members of family was 232 years. 3 yrs later, aged 60 yrs died, the next day baby was born in family. 3 yrs again later. 60 yrs aged person died & a baby was born the next day find the present avg age of this family.

a) 22
b) 23
c) 24
d) 25.

232
↓ 8 yr.
256 − 60
↓ 3 yr
256 − 60 + 24 − 60
↓ 4 yr.
256 − 60 + 24 − 60 + 32
↓
$\frac{192}{8} = 24$

Old avg = 29
profit = 8 × 10 = 80 yrs.
loss = 60 × 2 = 120 yrs.
net = abs (120 − 80) = 40 (loss).

avg loss = $\frac{40}{8}$ = 5.

new avg = 29 − 5 = 24

* Avg. marks of class 97 students is 51, when new student added is his score is also added, the new avg. is 51.5.

97 × 51 = 4947
98 × 51.5 = 5047
100 answer

if new student scores exactly avg. marks, then the avg. remains constant.
But if he scores more, Avg ↑ses.

ans = Avg + $\frac{98}{2}$ = 51 + 49 = 100

$NV = OA \pm (n \pm 1)(D)$

$= 51 + (97+1)(0.5)$

$= 51 + 49$

$= 100$

* Avg. age of class of 39 students was 9 yrs, when teachers is also added avg. becomes 9.5.

$NV = 9 + (39+1)(1/2)$

$= 9 + (40) \times (0.5)$

$= 9 + 20$

$= 29$

* After 37 matches players batting avg. was x, in next match she scores 162 & gets out, ↑ing the avg by 3, find her batting avg.

$162 = x + (37+1)(3)$

$x = 48 \rightarrow$ OA

New Avg = $48 + 3 = 51$ (Answer)

Puzzle

(Imp) * 5 People lives in house.

1 H, 1 W, 3 C

Hint 1) ages of child → product = 36

2) Sum of ~~...~~ of child ages = House no.

3) Eldest child can play the piano very well.

36 →	36	1	1	→ 38
	18	2	1	→ 21
	12	3	1	→ 16
	6	2	3	→ 11
	4	3	3	→ 10
	~~2~~	~~3~~	~~6~~	
	9	4	1	→ 14
	9	2	2	→ 13
	6	6	1	→ 13

} we cannot decide which to choose.

So, 3rd hint let's us know, the eldest son should be there, i.e. unique

So, our answer is (9, 2, 2)

→ 20 kg rice @ 62rs/kg is mixed with 10 kg rice @ 56/kg

Group Avg: $\frac{A_1 N_1 + A_2 N_2}{N_1 + N_2}$ $\quad \frac{20 \times 62 + 10 \times 56}{20 + 10} = \frac{1240 + 560}{30}$

$= \frac{180\cancel{0}}{3\cancel{0}} = 60$

* In what ratio shall we mix rice 62/kg with 56/kg to get a mixture of 60/kg.

62 56

60

4 2

ratio = 2 : 1 (4 : 2)

* What ratio should we mix a 61% acid sol^n to a 97% acid sol^n to get an 88% sol^n

61 97

88

9 27

9 : 27

Ratio : 1 : 3

* A person gets a bike & helmet for a total of 13K rs, he sells the bike @ 31% profit & helmet @ 8% loss, there by making a net profit of 19%. find cost price of bike.

31 -8

19

27 12

9 : 4

* In a company the avg. salary of 29 officers is 35 k rs whereas the avg. of non-officers is 11 k rs. if avg. of entire company is 15 k rs, then find no. of non officers, total no. of employees.

35 k   11 k

15 k

4   :   20

$\boxed{1:5}$

no. of non officers = 29 × 5
= 145

145 + 29 = 174

* Mixtures

mixtures

Two containers contains A & B contains milk & water in ratios 4:1 & 1:3 respectively. In what ratio should we mix A & B to get mixture C which contains milk & water in ratio 3:2

	M	W
A	4 :	1
B	1 :	3
C	3 :	2

Ratio of milk :-

(A) $4/5$ $1/4$ (B)

$3/5$ (C)

$\frac{7}{20}$ $\frac{1}{5} \times \frac{4}{4} = \frac{4}{20}$

$\boxed{7:4}$

imp) * A container contains 5 part syrup & 3 part water (a liquid i.e.). What part ~~must be~~ of this liquid must be removed & replaced by water, so that syrup & water are finally present in equal quantities.

	S	W
part	5	3
replaced	1 :	1

(original) $\frac{3}{8}$ (replaced) 1

$\frac{1}{2}$ (final)

$\frac{1}{2}$ $\frac{1}{8}$

Ratio = $\frac{1}{2} : \frac{1}{8}$

$= 4 : 1$

replaced part.

So, our answer is $\frac{1}{4+1} = \frac{1}{5}$

OR

8L mixture

5L Syrup → 4L Syrup

3L Water → 4L Water

Final → 4L Syrup, 4L Water.

we have to remove 1 L syrup & it is not added at last, so the part to be removed is our answer, i.e. 1 L out of 5 L.

$\frac{1}{5}$ Answer.

V.V. Imp) Container A contains 100 L of pure milk & B contains 100 L of pure water. Two buckets of milk is removed from A & poured into B. By the same 2 buckets of mix. in B is removed & poured back into A. The %age of water in A be x and y be the %age of milk in container B. Then ,

a) $x > y$

b) $x < y$

c) $x = y$

d) depends on capacity of Bucket.

$A \xrightarrow{\text{20L Milk}} B$

A	B
100L − 20L	100L + 20L
Milk	Water

⇓

$A \xleftarrow{\text{20L mix.}} B$

A	B
80L + 20L	120L − 20L
~~Milk~~	~~Water + milk~~

⇓

A	B
100L	100L
Milk + Water	Water + Milk
~~(77.33)~~ ~~(16.~~	

→ If we have 10 coins, having 8 Heads & 2 Tails, and we have to create 2 groups consisting of same numbers of Tails & can consist of any number of Heads. How to do this ?

So, there can be 3 possibilities.

8 H	2T
7H 1T	1H 1T
6H 2T	2H

(Now flip the coins on right)

0 Tails	8H.	2H.	0 Tails
	7H 1T	1T 1H.	
	6H 2T	2T	

* Mean

Arithmetic Mean, Geometric Mean, Harmonic Mean

$n = 2.$

$$AM = \frac{x_1 + x_2 + x_3 + \cdots + x_n}{n} \qquad \frac{x_1 + x_2}{2}$$

$$GM = \sqrt[n]{x_1 \times x_2 \times x_3 \times \cdots x_n} \qquad \sqrt{x_1 x_2}$$

$$HM = \frac{n}{1/x_1 + 1/x_2 + \cdots + 1/x_n} \qquad \frac{2xy}{x+y}$$

$$\boxed{AM \geq GM \geq HM}$$

$$\boxed{GM^2 = AM \times HM}$$

→ When going from Pune → Indore, A's speed is 40 Km/h, while coming back his speed is 60 Km/h. find his avg. speed throughout the journey. (distance is same)

→ While going from Pune → nashik. B's speed is 40 km/h for the first 2 hrs & 60 km/h for next 2 hrs find his avg. speed throughout the journey. (time is same)

$ If distance is same, avg. speed = $\underline{HM}$ $= \frac{2 \times 60 \times 40}{100} = \frac{4800}{100}$

$= 48$

If time is same avg. speed = $\underline{AM}$ $= \frac{40+60}{2} = \underline{50}$

__/__/____

* Car owner buys petrol at Rs. 7.5, 8, 8.5 per litre for three successive years. what exactly is the avg. cost per litre of petrol if he spends RS. 4000 each year.

→ This is the classic Harmonic mean problem.

→ had the consumption been same, we would've used Arithmetic mean.

* While going from pune → Bombay, A's speed is 20 km/h, while coming back A's speed is x km/h. find x, if her avg. speed is 40 km/h.

(H.W)

(Imp) * 25 Horses, 5 Lane Track, fastest 3, find minimum no. of races needed to get the fastest 3 horses.

	C1	C2	C3	C4	C5
R1	1	2	3	4	5
R2	6	7	8	9	10
R3	11	12	13	14	15
R4	16	17	18	19	20
R5	21	22	23	24	25

lets say 5^{th} member of each row won the race; & 4^{th} member of each row came 2^{nd} & similarly 3^{rd} member of each row came 3^{rd}.

_/_/_

Work and Time

* 2 workers can finish a job in 20 & 30 days respectively.

LCM (20, 30) = 60 units of work.

$\frac{60}{20} = 3$ efficiency of A

$\frac{60}{30} = 2$ efficiency of B

In how many days job will be done, if a & b work together.

efficiency(A + B) = 2 + 3 = 5

work = 60

Days = 60 / 5 = 12 days.

* A & B start working together after how many days should A leave the job. if the total work is completed in 24 days by B

24 × 2 = 48 units done by B.

remaining = $\frac{12}{3}$ = 4 → no. of days after A left.

* A & B start together, after how many days should B leave the job if total work is completed in 18 days by A.

18 × 3 = 54 units done

remaining = $\frac{6}{2}$ = 3 days.

~~After how many days~~

→ 3 workers A, B, C can finish a job in 12, 15, 20 days respectively.

e of A = 60/12 = 5
e of B = 60/15 = 4
e of C = 60/20 = 3

① in how many days will the job be done if A, B, C work together.

$5+4+3 = 12$. 60 units $/12$ = $\boxed{5 \text{ days}}$

② after the job was done they were paid a total of 3600 rs, find the share of A.

$\frac{5}{\cancel{12}} \times \cancel{3600}^{3} = 1500$ Rs.

③ A, B, C work together for 2 days. now C leaves the job, in how many days will A & B together finish the remaining work.

for 2 days, $12 \times 2 = 24$ units of work done.

Remaining : $60 - 24 = 36$ units.

(A+B) efficiency = 9

∴ days = $36/9$ = $\boxed{4 \text{ days}}$

(4) A works alone for 3 days, then B works alone for 6 days. in how many days C finish the remaining work.

A → 5 × 3 = 15

B → 4 × 6 = 24

39

remaining : 60 - 39 = 21

(5) A, B, C works together for 3 days, now B leaves the job, in how many days will A & C will complete the work if they work on alternate days (Starting with A)

6 days

(6) A, B, C work together, now B leaves job. find the exact time taken to finish the remaining days if A & C work on alternate days (Starting with A)

(12 days remaining)

A → 12 - 5 = 7.

C → 7 - 3 = 4

A → $\frac{4}{5}$

∴ (2 + 4/5) days

(7) In que. (6), what if C & A work alternatively starting with C

C → 12 - 3 = 9

A → 9 - 5 = 4

C → 4 - 3 = 1

$(3 + 1/5) = \underline{3.2}$

* 2 workers A & B can finish job in 16 days & 12 days resp

$A \rightarrow 48/16 = 3$

$B \rightarrow 48/12 = 4$

find exact time taken to finish the job if A & B work on alt. days starting with A.

combined eff. $= 3+4 = 7$ units. (every 2 days)

so, $42/7 = 6 \rightarrow 6 \times 2 = 12$ days.

remaining units $= 6$ units.

Now. $A \rightarrow 6 - 3 = 3 \rightarrow 13$

$B \rightarrow 3/4. \rightarrow$ 13 3/4 days

* 2 workers A & B together can finish a job in 12 days. while B & C together takes 16 days while A & C together take 24 days. In how many days will the job be done by C alone?

$a + c = 48/12 = 4$ LCM : 48

$b + c = 48/16 = 3$

$a + c = 48/24 = 2.$

$2(a+b+c) = 9$

$\boxed{a+b+c = 4.5}$

$a = 1.5$

$b = 2.5$

$c = 0.5$

_/_/_

* Efficiency $\propto \frac{1}{\text{Time}}$ (inversely proportional)

* Efficiency $\propto$ work (directly proportional)

* A is twice as efficient as B, if they work alone, A takes 11 days less than B to finish a job. In how many days B can finish it alone.

$x = 11$ days.

$e(A) = 2x$.

ans $= 2 \times 11 = 22$ days.

* A is twice efficient as B. So, if they can start working together they can finish a job in 30 days. In how many days can A finish it Alone?

* Puzzle

* 9 identical balls in front & 1 odd ball among this 9 which is heavier than other 8 balls. (weighing machinge is given & find minimum number of times needed to machine)

9 → A, B, C

A, B → Check, if same → move to C

if not same → the group with heavy will be checked

(Cybage Puzzle) * 12 identical looking Tennis Balls, given ~~weight~~ weighing machine and one ball is heavy & (maximum no. of times we can use machine is 3).

distribution → 12 → 1,2,3,4 | 5,6,7,8 | 9,10,11,12

* A is 30% more efficient than B, if they work together they can finish the job in 13 days, in how many days can A finish the work if he works alone.

A → 1.3

B → 1

$\left.\begin{array}{l} A \to 1.3 \\ B \to 1 \end{array}\right\} \dfrac{2.3 \times \cancel{13}}{\cancel{13}} \to$ ~~203~~ 23

_/_/_

* 2 grape crushers que. ans. (App Dynamic Ques)

2 combined takes 4 days

& 1/2 Stock by one crusher & 1/2 stock by other crusher takes 9 days.

→ lets take lcm (9,4) = 36 stock, we have to crush

now,

$$a + b = 9 \text{ kg}$$

stock ← $\frac{18}{a} + \frac{18}{b} = 9$ days

now solve for efficiency of a & b

_/_/_

* Pipes & Streams

An inlet pipe can fill the tank in 20 hours.
an outlet pipe can empty it in 24 hrs.
in how many hours will the tank be full if both work together?

(Give - sign to outlet & + sign to inlet)

I → 20 hrs
O → 24 hrs
lcm (20, 24) → 120

$\frac{120}{20} = +6$, $\frac{120}{24} = -5$ → $6 + (-5) = 1$

→ Two inlet pipes A & B can fill a tank in 8 & 12 hrs respectively while an outlet pipe C can empty the same tank in 6 hours. At 1 pm only pipe A was opened. At 3 pm. B was also opened & At 5 pm C was also opened. At what exact time will the time be filled.

A → 24/8 = 3 → 1 pm
B → 24/12 = 2 → 3 pm
C → 24/6 = -4 → 5 pm

6L
10L

lcm (8, 12, 6) = 24

Till 5pm = 16 L is there in tank

Total combined flow each hour = 3 + 2 + (-4)
= 1 L/hr

Still remaining = 8 L.

So from 5 pm we'll require 8 hr to fill 8 L. so,
5 + 8 = 1 pm

11 12 1
8 hr

__/__/____

(Imp) * An inlet pipe A can fill the tank in 15 h & B can empty the tank in 25 h. Find exact time to fill the tank if A & B works alternatively.

$lcm(15, 25) = 75$

$A = \frac{75}{15} = +5L/hr$

$B = \frac{75}{25} = -3L/hr$

Combined O/p = $+2L / 2hr$

$70L \rightarrow 70h$

5L remaining → So in next hr when, A is on it'll fill 5 litre & the tank will be full in next hr.

$70 + 1 = 71$ hours

* 11 Que

$I \rightarrow 120/24 = 5$

$lcm(24, 40) = 120$

$I - L = 120/40 = 3$

inlet, leak

$L = 5 - 3 = -2$

So, to empty $120 \rightarrow 120/2 = 60$ hours.

Chain Rule

$W \propto T$, $W \propto E$, $E \propto \frac{1}{T}$, $D \propto H$, $W \propto \frac{1}{T}$

W → work, T → Time, W → work, E → Efficiency, T → Time, D → Days, H → Hours, W → workers

→ For direct proportion, keep ratio direct.

→ For inverse proportion, take the inverse of 2^{nd} ratio.

Que → 24 engineers finish proj. in 12 days, in how many days can 32 engineers finish the same project,

Relation is inverse, i.e. if we have more engineers, we will take less time.

$24 \rightarrow 12$

$32 \rightarrow x$

$$\frac{24}{32} = \frac{x}{12} \Rightarrow \frac{24 \times 12}{32} = x$$

$x = 9$

* A merchant employed a person, and promised him an annual salary of 9000 cash & gold chain, however the servant work for only 9 month. he got 6500 cash & a gold chain. find the value of gold chain.

$12 \rightarrow 9000 + G$

$9 \rightarrow 6500 + G$

$$\frac{\cancel{12}^{4}}{\cancel{9}_{3}} = \frac{9000+G}{6500+G}$$

$$\boxed{G = 1000}$$

* 24 painters work 8 hrs/day can paint 1 building in 5 days in how many days can 45 painters working 4 hrs a day paint 3 such buildings.

P	H	D	B
24	8	5	1
45	4	x	3

$$(i)\ \frac{\cancel{45}^{\cancel{15}\,9}}{\cancel{24}_{\cancel{8}\,2}} \times \frac{1}{\cancel{3}_{1}} \times \frac{\cancel{4}^{1}}{8} = \frac{\cancel{5}}{x}$$

$$\boxed{x = 16}$$

$P \propto \frac{1}{D}$ & $D \propto B$

& $H \propto \frac{1}{D}$

* 17 Que

M	D	A
175	12	1/4
175 + x	7	3/4

how many more men required

* 6 men or 8 women can finish job in 33 days. in how many days 15 men & 4 women finish a similar job.

Step 1:- Compare → 6M = 8W → $\boxed{M = 4/3\, W}$

Step 2:- Convert

(convert in less efficient worker/entity)

$15\,M = \frac{4}{3} W \times 15 = 20\,W + 4\,W$

Total women working = 24 W.

W	D
8	33
24	x

Inverse relation:- $\frac{\cancel{8}^{1}}{\cancel{24}_{3}} = \frac{x}{33}$

$x = 33/3 = 11$

$\boxed{x = 11 \text{ days}}$

* 12 men & 16 boys can construct a wall in 5 days. 13 men & 24 boys construct the similar wall in 4 days. in how many days 8 mens & 9 boys can construct a similar wall.

$$(12M + 16B) \times 5d = (13M + 24B) \times 4d$$

$$60M + 80B = 52M + 96B$$

$$8M = 16B$$

$$\boxed{1M = 2B}$$

$$60 \times 2B + 80B = \text{Total work}$$

$$120B + 80B = 200 \text{ Boys}$$

$$\text{Time} = \frac{200B}{8M + 9B}$$

* 7 inch gold bar, 1 day work = 1 inch gold bar,

1	2	4

Day 1 → 1 inch gold given. [2 | 4]

Day 2 → (worker returns 1 inch) & we give 2 inch bar. [1 | 4]

Day 3 → we give him 1 inch gold [4]

Day 4 → (worker returns 3 inch gold) & we give 4 inch gold bar. [1 | 2]

Similarly thinks in powers of 2 or 3 or 5 in such problems

40 — 1, 3, 9, 27

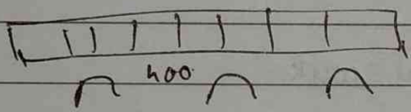
Speed, Time, Distance

1 Km/h = $\frac{5}{18}$ m/s

* d = 396 km   T = 11 hr.

S = 36 Km/hr. = 10 m/s

* Bridge = 400 m   * Train length = 360 m



Total dist = 760 m

Speed = 36 Km/h = 10 m/s $\frac{\cancel{36}^{2} \times 5}{\cancel{18}_{1}}$

$\boxed{\text{Time} = 76\ s}$

* Whenever, two moving objects is given, assume slower objects is stationary & then use concept of relative speed

Same Direction
Speed = $x - y$

Opp. Direction
Speed = $x + y$

* P (Point) L (lengthy) S (Stationary) M (moving) → filter

_/_/_

* Train overtakes 2 persons, who are walking with speeds 2 & 4 km/h in 9 & 10 seconds respectively. find length of train.

a) 36

b) 50

c) 54

d) 72

$\frac{10}{18}$ & $\frac{20}{18}$ $\Rightarrow$ $\frac{5}{9}$, $\frac{10}{9}$

$d = (s - 5/9) \times 9$ | $d = (s - 10/9) \times 10$

$d = 9s - 5$ | $d = 10s - \frac{100}{9}$

$9s - 5 = 10s - \frac{100}{9}$

$\frac{100}{9} - 5 = s$

$\boxed{s = \frac{55}{9}}$

$d = 9 \times \frac{55}{9} - 5$

$\boxed{d = 50\ m}$

* Train Overtakes 2 persons walking at 4.5 & 5.4 km/hr in 8.4 & 8.5 sec resp. find speed of train.

a) 66

b) 75

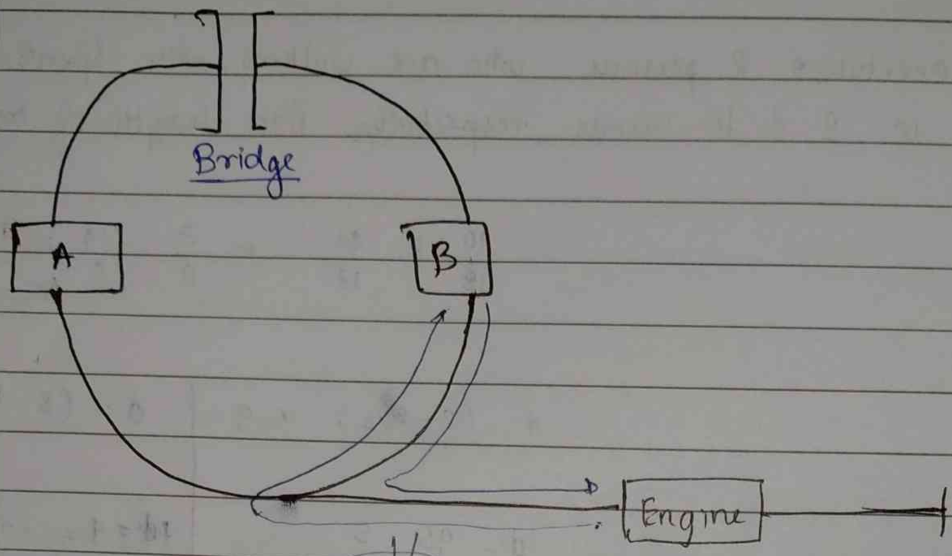
c) 78

d) 81

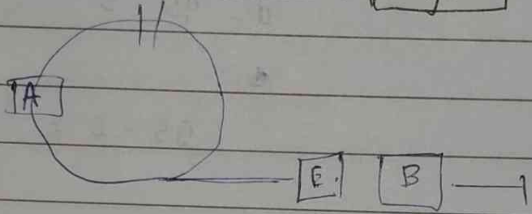
Only multiple of 9.

(∵ most of the time speed is a multiple of 9)

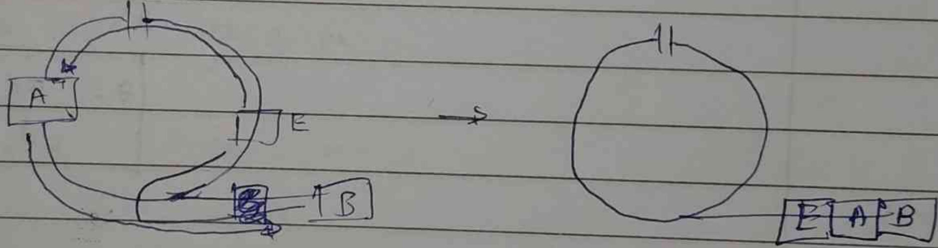
*



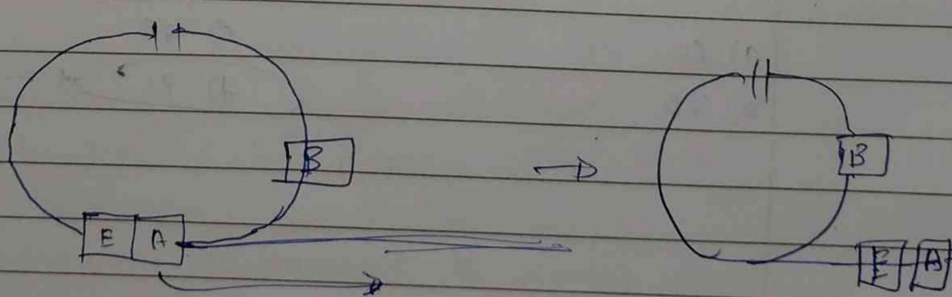
(1)



(2)



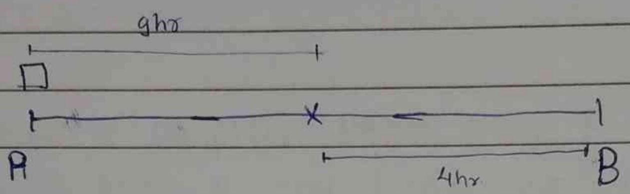
(3) Pull all 2 bogeys out & leave B at first section



(4) leave A in garage and now attach B to Engine & take back to A

Imp | Train A leaves Pune for Delhi at 72 km/hr (20 m/s) at the same time Train B leaves Delhi for Pune at x km/hr. After crossing each other Train reaches delhi in next 4 hrs while train B reaches pune in next 9 hrs. find x.

a) 32 b) 40 c) 48 d) 56



Perfect Sq. Time

★★★★ $\boxed{\frac{S1}{S2} = \sqrt{\frac{T_2}{T_1}}}$

$$\frac{72}{x} = \sqrt{\frac{9}{4}}$$

$$\frac{72}{x} = \frac{3}{2}$$

$$x = \frac{\overset{24}{\cancel{72}} \times 2}{\cancel{3}}$$

$$\boxed{x = 48}$$

VImp * A & B are walking towards each other on 120 km long road with speeds 5 & 3 km/h ($5\times\cancel{10}/\cancel{18}$ m/s, 5/6 m/s). At the same time a Bird sitting on A's hat starts flying towards B, touches B' hat & fly's back to A & so on. ~~By~~ ~~Speed~~ till ~~they~~ A & B meet.

If the speed of Bird is (5m/s) 18 Km/hr. Find the distance travelled by bird.

~~Bird~~ ~~8~~5 3

A ⟶ ⟵ B

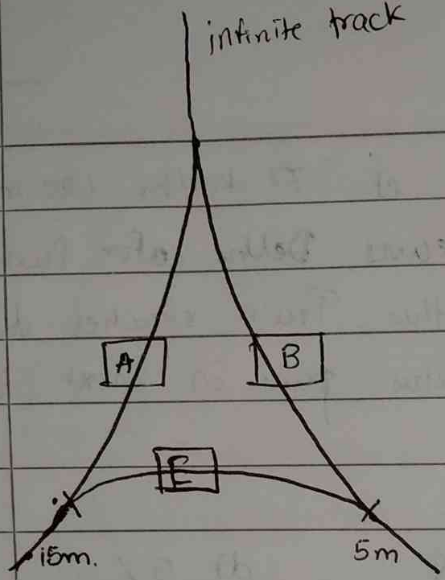
Total time = $\frac{120}{8}$

= 15 h

So, total dist = 18 × 15 = 270

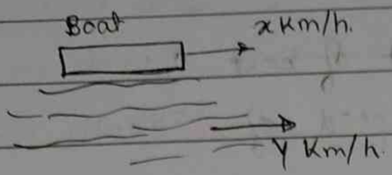
___/___/___

*



* Swap B and A & Engine should be at original position.

* Boats and Streams



Downstream = $(x + y)$ km/h

Upstream = $(x - y)$ km/h

* Boat is always >= river speed
$(B >= R)$

* A boy rows to a place 60 km away & comes back, find total time taken if speed of boat & river are 10 & 5 km/hr respectively.

$$\frac{60}{10+5} + \frac{60}{10-5} = 16\text{ h.}$$

* A rows to a place 55 km away & comes back taking a total of 16 hours find the speed of boat if speed of river is 3 k/h

$$\frac{55}{(x-3)} + \frac{55}{(x+3)} = 16$$

5   11

$$55(x+3) + 55(x-3) = 16(x^2 - 9)$$

$x - 3 = 5 = \boxed{x = 8}$

* Manoj rows to a place 96 km away & comes back taking a total of 20 hours find the speed of boat if speed of river is 2 km/h.

$$\frac{96}{(x+2)} + \frac{96}{(x-2)} = 20$$

12   8

$x + 2 = 12 \rightarrow \boxed{x = 10}$

* A man rows 48 km downstream & Come back taking a total of 14 hrs. If ratio of ~~Speed of boat~~ downstream speed to upstream speed is 4:3, then find the speed of Stream.

$$\frac{D}{U} = \frac{4}{3} = \frac{B+R}{B-R}$$

* A man travels 30 upstream & 34 down stream in 10 hrs & can also travel 40 km upstream & 55 km downstream in 13 hrs. Find speed of boat in still water.

$$\frac{30}{x-y} + \frac{44}{x+y} = 10$$

$x+y \rightarrow$ assume $(x+y) = 11$

$\frac{44}{11} = 4$ $\quad 10 - 4 = 6$

$\frac{30}{6} = x - y$

$\boxed{x - y = 5}$

$\boxed{x + y = 11}$

$\boxed{x = 8}$

$\boxed{y = 3}$

$$\frac{40}{x-y} + \frac{55}{x+y} = 13$$

* $\frac{\text{Upstream time}}{\text{Downstream time}} = \frac{4 \text{ hr } 48 \text{ min}}{4 \text{ hr}}$

$= \frac{\cancel{528}^{11} \text{ min}}{\cancel{240}_{5} \text{ min}} = \frac{11}{5}$

But $T \propto \frac{1}{S}$

So, $\frac{UT}{DT} = \frac{\text{Downstream Speed}}{\text{Upstream Speed}} \frac{(B+R)}{(B-R)} = \frac{11}{5}$

$\boxed{\begin{array}{l} B = 8 \\ R = 3 \end{array}}$

_/_/_

P&C

* Counting Principles

* If job 1 can be performed in x no. of ways & job 2 in y no. of ways then the no of ways in which we can perform

J1 OR J2 is $x+y$

J1 AND J2 is xy

* White shirts = 3
  Black —"— = 2

Choices of shirt : 3 + 2 = 5
White OR Black

* A has 5 Tshirts & 3 Trousers in how many ways can he dress.

Choice of TShirt $= {}^5C_1 = 5$

choice of Pants $= {}^3C_1 = 3$

ways $= 5 \times 3 = 15$

* ways to arrange 5 students in 2 chairs.

$\underline{5} \times \underline{4} = 20$

* 5 people, 3 chair → $5 \times 4 \times 3 = \underline{\underline{60}}$

* 5 people 5 Chair → 5!

Rule 1 :- n distinct objects can be arranged among themselves in n! ways.

* Ways to Rearrange "WATER", if

1. Vowels are always together
2. — " — never — " —
3. consonants are always together
4. — " — never — " —

1. — — — — —

AE α WTR

4 entities WTRα.

4! × 2!

internal arrangement of A & E

2. Total − Always
= 120 − 48
= 72

3. WTR A E
1 2 3

3! × 3! = 36

4. Total − Always =
120 − 36
84

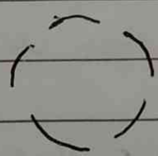
* How many arrangements of the word google are possible.

* Out of n' objects, if P obj are of type 1 & q obj of type 2 & r obj of type 3.
Then total no. of arrangements is

$$\text{Total} = \frac{n!}{P!\,q!\,r!}$$

$$G\ O\ O\ G\ L\ E \rightarrow \frac{6!}{2!\,2!\,1!\,1!} =$$

* In how many ways we can arrange 5 students into 5 chairs kept in a circle.



$$(4 \times 3 \times 2 \times 1) = \underline{\underline{4!}}$$

for stud 1 :- he has only 1 choice, as the position of each chair is not determined.
for stud 2 :- he will now have 4 choices.
for stud 3 :- —— " —— 3 —— " ——
for stud 4 :
⋮

* N people can be arranged in a circle in (N-1!) ways as the first person has only 1 choice.

_/_/_

* Permutation (Arrangement) (ORDER Important)

* n obj can be arranged in r places in ${}^nP_r$ ways

$$ {}^nP_r = \frac{n!}{(n-r)!} $$

* Combination (Selection) (Order Unimportant)

→ r objects can be selected from n objects in ${}^nC_r$ ways

$$ {}^nC_r = \frac{n!}{(n-r)!\, r!} $$

→ OR ↓ + AND ↓ × OI (order imp) ↓ P ~~order~~ OU (order unimp) ↓ C

OR	AND	OI (order imp)	OU (order unimp)
↓	↓	↓	↓
+	×	P	C

* In how many ways of 5 students from a group of 4 boys & 5 girls if comittee comprises of

① exactly 2 boys $\quad {}^4C_2 \times {}^5C_3$

② exactly 4 girls $\quad {}^4C_1 \times {}^5C_4$

③ no boys $\quad : {}^5C_5$

⑤ Atleast 3 girls. $\quad {}^4C_2 \times {}^5C_3 + {}^4C_1 \times {}^5C_4 + {}^4C_0 \times {}^5C_1$

* How many straight lines we can draw, on a plane having 10 non-colinear points.

$$ {}^{10}C_2 = \frac{10 \times 9}{2} = 45 $$

No. of Triangles = ${}^{10}C_3$ (as order is unimp)

* How many 5 digit no. can we make using the digits {1, 2, 3, 4, 5} (repetition not allowed) that are divisible by.

divisible by

① 2 → $\underbrace{\_ \ \_ \ \_ \ \_}_{4!} \,|\, \underset{2}{\_}$ → $4! \times 2$

② 4 → $\underbrace{\_ \ \_ \ \_}_{3!} \,|\, \underbrace{\_ \ \_}_{4}$ (last two → 12, 24, 32, 52) → $3! \times 4$

③ 5 → $4!$

* Dictionary Order

* If all the arrangements of the word "WATER" are arranged in dictionary then find the rank of "WATER"

A _ _ _ _ → 4!

E _ _ _ _ → 4!

T _ _ _ _ → 4!

R _ _ _ _ → 4!

W A _ _ _ → 3!

W A T _ _ → 2!

W A T E _ → 1!

* MUCH → Dictionary Rank

C _ _ _ → 3! (1 – 6)

H _ _ _ → 3! (7 – 12)

M _ _ _ (13 – 18)

C H U

Probability

→ Coins

For n coins , 2^n outcomes are there. (coins = C) order unimp

* What is probability of getting

1. exactly 3H → ${}^6C_3/64$ →
2. atleast 5H → $\frac{{}^6C_5 + {}^6C_6}{64}$
3. atmost 2H.

$$\frac{{}^6C_0 + {}^6C_1 + {}^6C_2}{64}$$

→ Cards (C) order unimp

52 Cards

26 Black → ♡ 13, ◇ 13

26 Red → ♧ 13, ♤ 13

* There are 12 face cards
* There are 40 non-face cards

* Dice (C)

P(even num) = 3/6
P(perfect sq) = 2/6

for n dice → 6^n outcomes

what is P of getting if 2 dice rolled :-

min sum → 2

max sum → 12

① Sum as 7

② Sum is prime

$\frac{15}{36}$ $\frac{7}{36}$

Sum	Ways	
2	1	1
3	2	2
4	3	
5	4	4
6	5	
7	6	6
8	5	
9	4	
10	3	
11	2	2
12	1	

(for 2 dice)

* Bag 1 contains 5 orange and 3 Green balls, bag 2 contains 7 Orange & 1 green balls.
A ball is selected from bag 1, put into bag 2, now is ball i selected from bag 2, what is the propability that it is an orange ball.

B1 → 5 O 3 G

B2 → 7 O 1 G

G and O OR O and O

$\frac{3}{8} \times \frac{7}{9} + \frac{5}{8} \times \frac{8}{9} = \frac{61}{72}$

__/__/__

* Probabilities that A, B, C can solve a problem R are 1/2, 1/3 and 1/4 respectively. what is the prob. that the problem is solved?

A	B	C
$\frac{1}{2}$	$\frac{1}{3}$	$\frac{1}{4}$

Prob of solving $= 1 - \left(\text{prob of not solving}\right)$

$$= 1 - \left(\frac{1}{\cancel{2}} \times \frac{\cancel{2}}{\cancel{3}} \times \frac{\cancel{3}}{4}\right)$$

$$= 1 - \left(\frac{1}{4}\right)$$

$$= \frac{3}{4}$$

* A no. is selected from $1 \rightarrow 60$. what is the prob. that the given no. is a multiple of 6 or 8 or both

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$= \frac{P(6) + P(8) - P(6 \cap 8)}{\cancel{60}}$$

$$= \frac{10 + 7 - 2}{60}$$

$$= \frac{\cancel{15}}{\cancel{60}} \; \frac{1}{4}$$

$$= \frac{1}{4}$$

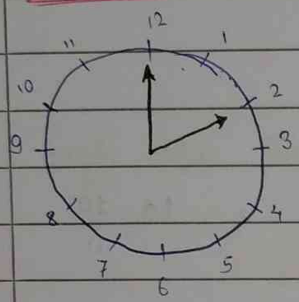
* Puzzle

8 Battery Cells

4 Good — 4 Bad

minimal pairings to start the Device with two batteries.

* Clocks



* Clock = 60 min spaces
* Speed of min hand = 60 m.s./hr

  = 1 m.s/min.

* Speed of hr hand = 5 m.s./hr

  = $1/12$ m.s./min

* Relative Speed of min hand wrt hr hand = $1 - \frac{1}{12}$ m.s/min

  = $\frac{11}{12}$ ms/min

* The two hands coincide 11 times in 12hr. and i.e.. 22 times in 24 hours
* The two hands are in opp. direction in 11 times in 12 hr i.e. 22 times in 24 hours
* The two hands make 22 ~~angle~~ right angles in a 12 hr period, i.e. 44 right angles in 24 hrs.

* In a day, how many times the hands are in a straight line.

→ 22 (when in opp. dir^n) + 22 (when coincide)
→ 44

* At what exact time b/w 2 & 3 will the two hands coincide.

dist. b/w min & hr hand = 10 m.s.

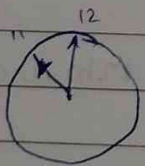
R.S of min hand = 11/12.

∴ time = $\frac{D}{S} = \frac{10}{11/12} = \frac{120}{11} = 10\frac{10}{11}$

Exact time $\boxed{2 : 10\ {}^{10}/_{11}}$

* $\boxed{\text{Whatever you see, multiply by } \frac{12}{11}}$

* At what exact time b/w 11 & 12 will the two hands coincide



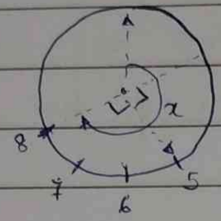
d = 55 $t = \cancel{55}^{5} \times \frac{12}{\cancel{11}} = 60$

11 : 60 → 12 : 00 or never

(If given in option)

* b/w 5:30 & 6 when will the two hands make a right angle

Consider the distance from 12 always



$x = 40$

$$\therefore T = \frac{d}{s} = \frac{40}{11/12} = \frac{40 \times 12}{11} = \frac{480}{11} = 43\frac{7}{11}$$

* Angle b/w Two hands at time h:m is

$$|30^\circ H - 5.5^\circ M| \rightarrow \text{absolute value}$$

* Find reflex angle b/w two hands at 10:25

reflex angle means : $360^\circ > \text{angle} > 180^\circ$

$= 360 - |30 \times 10 - 5.5 \times 25|$

$= 360 - |162.5|$

$= \underline{197.5}$

* A pendulum clock takes 7 seconds to strike 7, how many secs. will it take to strike 10.

6 spacing 7 seconds 10.5

9 spacing x sec.

$\cancel{6}^{2} x = \cancel{9}^{3} \times 7 \rightarrow x = \frac{21}{2} = \underline{10.5}$

* a clock uses 4 mins daily, after how many days will this clock show the correct time.

* 12 Balls Puzzle

A	B	C
1 2 3 4	5 6 7 8	9 10 11 12

[1 2 3 4] — [5 6 7 8]

choose → 2 from A & 1 from B

[1, 2, 5] — [3 6 9]

	Defective balls
same ⟷	4, 7, 8
high ↑ ↓ low	1, 2, 6
low ↓ ↑ high	5, 3

Now from defective balls, from same group & then compare

[7] — [8]

① ↑ ↓ → 8 heavy ~~&~~ or 7 ~~lighter~~ light.

② ⟷ → 4 defective ball.

[1] — [2]

① ↑ ↓ → 1 light or 2 heavy

② ⟷ → 6 defective

* Mouse has 5 cages

1 2 3 4 5

in extreme he can switch in only one place where in 1 -> 2 & 5 -> 4. , but if in

2 3 4 2 3 4 (or 4 3 2, ~~or~~ 4 3 2 or 3 2 4 or 3 2 4)

(Play with"

* Calendar

जो ~~दिन~~ divide by 7

→ 59 days ago → 59/7 = 3

Saturday - 3 days = wednesday

→ leap year = (year % 4 == 0 && year % 4 != 0) ||
(year % 400 == 0)

→ for a century it is divisible by 400
→ for a non-century it is divisible by 4.

→ if 29th nov 2025 is saturday, find the day on 29th nov 2029

in non leap year → 1 odd day added
in 1 leap year → 2 odd days added

NLY - 2025, 2027, 2026 → 3×1 = 3
LY - 2028 → 1×2 = 2
} 5 + ~~Saturday~~ = Thursday.

* 15 Aug 1917

→ 7 leap years

15 Aug 1947 Fri

→ 8 leap years

15 Aug 1977

$30 + 7 = 37 \% 7 = 2$ Fri - 2 = Wed

$30 + 8 = 38 \% 7 = 3$ fri + 3 = Mon.

* $4x + 1 \rightarrow 6y$

$\left.\begin{array}{l} 4x+2 \\ 4x+3 \end{array}\right\} 11y$

$4x$ (LY) $\rightarrow 28y$

* The last day of a century can only be Friday (5), wednes (3), Sunday (0) or Monday (1)

and never, Tuesday, thursday & Saturday

Percentage

50 %	1/2
25 %	1/4
10 %	1/10
12.5 %	1/8
75 %	3/4
80 %	4/5
83.33 %	1/3
66.66 %	2/3
16 . 66 %	1/6
83 . 66 %	5/6
120 %	6/5
125 %	5/4
133 . 33 %	4/3

<u>Standard</u> % → frac.

1/7	14.28 %
1/11	9 . 09
1/12	8.33
1/15	6.66
1/9	11.11
1/16	6.25

* A B C D are such that.

$\frac{A}{B} = \frac{5}{6}$, $\frac{B}{C} = \frac{3}{4}$, $\frac{C}{D} = \frac{1}{3}$

find A if total = 12000.

$\frac{5}{\cancel{6}} \times \frac{\cancel{8}}{\cancel{4}} \times \frac{1}{\cancel{3}} \times \cancel{12000}^{\cancel{2000}\ 500} = \underline{2500}$

* Round of when needed 594 → 600

* $100 \xrightarrow{\uparrow 20\%} 120 \xrightarrow{\downarrow 20\%} 120 - 24 = 96$

Net % = 4% loss

* $100 \xrightarrow{\uparrow 30\%} 130 \xrightarrow{\downarrow 30\%} 130 - 39 = 91$

$\frac{100 - 9}{\cancel{100}} \times \cancel{100} = 9\%$ loss

* $100 \xrightarrow{\downarrow 30\%} 70 \xrightarrow{\uparrow 30\%} 91$

first ↑ & ↓ or ↓ & ↑ → output will be same
So, o/p wont wont change

* If a no. is increased and decreased by $x\%$, then there is loss of $\left[\frac{x}{10}\right]^2$

* $E = P \times C$

Expenditure = Price × Consumption

* Price of mango ↑ses by 30%, while consumption ↓ses by 1
find % change in expenditure

assume e = 100 ⟶ Price ↑ 30% = 130

↓ Consump ↓ 10%.

130 − 10 = 117

net change = 117 − 100 = 17 %

* l & b of room ↑se 25% & 40% respectively but height is reduced by 20%. find % change in volume

100 → ~~25~~ 125 → 175 → 140

↓

40%

* Complementry pairs are those which negate the effects of each other.

for example

$100 \xrightarrow{50\%\uparrow se} 150$

33.33%

(33.33 & 50)

__/__/__

* Watermelon → in day 600 kg → 99% W 6kg pulp.

After keeping in sunlight.

new weight of watermelon is, if the new water% of watermelon is 98%.

600 kg

99% W	6 Kg	1%
$\underline{300} = \frac{600}{2}$ → 98% W	6 kg	2%

* If % share of constant entity (generally solid) is multiplied by k then the total weight will be divided by k.

* Fresh grapes contain 90% water & dried grapes has 20% water

	W	S
FG	90%	10%
DG	20%	80%

$\times 8 \longrightarrow k$

How many kg of dried grapes can we make from 40 kg of Fresh grapes?

Total → $\frac{40}{8} = 5$ kg

__/__/____

* In an election b/w 2 candidates every voter voted. 6% of voters pressed NOTA button, while 63% of total voters voted for candidate A who defeated candidate B by 1600 votes.

① Find total no. of voters ② # of votes recieved by b.

③ # of nota vote

100 %

A 63%

B 31%

NOTA 6%

32% = 1600

1% = $\frac{\cancel{1600}^{\cancel{100}50}}{\cancel{32}}$

1% = 50 voters

que solved

* Profit & loss

100

WD	SAV
50	50
30	20
10	10
90	80

Imp Terms

CP , SP , P , L

(cost price) (sell price) (profit) (loss)

$P/L\% = \frac{P/L}{C.P} \times 100$

__/__/__

(CP same)* A buys 2 books, 300 rs each, & sells 1^{st} book at 20 % Profit & 2nd one at 20 % loss, find her net profit or net loss %age.

<u>NO Profit</u> <u>NO Loss</u>.

as 20 % P & 20 % loss

so, total loss 0 %

(SP same)* B sells 2 Books at 800 rs each & on 1^{st} makes profit of 20 % on 2^{nd} B makes loss of 20 %. Find net loss/profit %age

If CP is same no profit no loss.

If SP is same, then there is always a loss of $\left[\frac{x}{10}\right]^2$.

i.e. $\left(\frac{20}{10}\right)^2 = 4\%$

* class of 200, 40 % women, $1/5^{th}$ of those become lecturers. If no. of men lecturers = 2 × women lect.
Calculate approximate % of men become lect

200 → 40 % → 80 → 1/5 → 16

men lect = 2 × 16
= 32

$\frac{32}{120} \times 100$

→ $20 < x < 30\%$

* Price of sugar ↓ses by 2.5%. a person can buy 9 kg more for 1260 rs.

If price ↑sed by 12.5 % how much sugar would he have bought for the same amount.

a) 288 kg c) 328 kg

b) 312 kg d) 326 kg

$$1260 = P \times C \qquad -①$$

$$1260 = \frac{39}{40} P \times (C+9) \qquad -②$$

$$1260 = \frac{9}{8} P \times x \qquad -③$$

① = ②

$$\cancel{P} \times C = \frac{39}{40} \cancel{P} (C+9)$$

$$C = \frac{39}{40} (C+9)$$

$$40C = 39C + 9 \times 39$$

$$C = 9 \times 39 = 351$$

$$\cancel{1260}^{340} = \frac{39P}{\cancel{40}} \times \cancel{360}^{9}$$

$$\frac{340}{39} = P$$

$$\frac{\cancel{1260}}{\cancel{340}} \times \frac{39}{\cancel{9}} \times 8 = x$$

$$x = 39 \times 8$$

$$\boxed{x = 312}$$

PTO

__/__/__

* Price ↑ses 33.33 %, A is able to buy 5 kg less, for a total of 240 rs. Find original Price/kg or sugar.

a) 10 b) 15

c) 12 d) 20

$240 = \frac{4}{3} P\ (C-5)$

$240 = P \times C$

$C = \frac{4}{3}\ (C-5)$

$3C = 4C - 20$

$\boxed{C = 20}$

$P = \frac{3}{\cancel{4}} \times \frac{\cancel{240}^{\cancel{60}\ 4}}{\cancel{15}\ 1}$

$\boxed{P = 12}$

* Carpenter sells 12 chairs at cost price of 14 chair find his profit or loss %age.

CP = ₹ 12

SP = ₹ 14

$\frac{14-12}{12} \cdot \frac{1}{6} \times 100 \Rightarrow 16.66\ \%$

_/_/_

* A fruit vendor apples at CP but uses a 800 gm weight instead of 1kg. Find the profit ?

$CP = SP = ₹1$

$$P = \frac{\text{Cheat Value}}{\text{Actual Value}} \times 100$$

$$= \frac{80\cancel{0}}{10\cancel{00}} \times 1\cancel{00}$$

$= 80\ \%$

* ~~[illegible]~~

A man sells 2 commodities for 4000 each neither losing nor gaining in the deal. if 1 sold at gain of 25%, then the other commodity was sold at a loss of ?

(Complimentry pairs negate each other)

CP	SP	P
80	100	25% (20)
120	100	

$$(120) \rightarrow \frac{20}{120} = \frac{1}{6} = 16.66\ \%$$

* Peter bought an item at 20% discount, sold at 40% profit at which he bought, what is % more than the original price

$$100 \xrightarrow{20\% \downarrow} 80 \xrightarrow{40\%} 32$$

$80 + 32 = 112 \longrightarrow \frac{112}{100}$

12 %

Simple & Compound Interest

$SI = \frac{PNR}{100}$ $A = P\left(1 + \frac{R}{100}\right)^N$

$A = P + I$ $CI = A - P$

→ ~~Ca~~ SI is a good example of an AP.
→ For CI the principle keeps on ↑ing every year, bcoz, we get interest on interest.
→ CI is good example of a GP.

* SI for 6 years = 30% P, then in how many years will the principle double.

$\frac{30}{6} = 5$ rs/year

$\frac{100}{5} = 20$ years

* A promises B, he will double his money in 21 days using compound interest. So after how many days B's ~~promises~~ principle will become ~~times~~

① 4 times → 42 days
② 8 times → 63 days

* The diff b/w CI & SI for 2 years @ 20% per annum is 84 Rs, find the principle.

- Difference = $\left(\frac{R}{100}\right)^2 P$

$$84 = \left(\frac{20}{100}\right)^2 P$$

$\boxed{P = 2100}$

* Total SI & total CI for 2 yrs on a P are 360 rs & 378 rs, find ROI.

	SI	CI
	360	378
1^{st} yr	180	180
2^{nd} yr	180	198

CI = SI for 1^{st} year every time

interest on interest

$198 - 180 = \underline{18\ \%}$

* Clocks (contd.)

→ For an incorrect watch, showing the correct time means gaining or losing 12 hours.

* Watch 1 gains 1 min daily & watch 2 loses 1 min daily. after how many days will these two watches show

① Same time → 1 min daily → 60 min → 60 days → so after 6 hours i.e. 60 × 6 = 360 days

② Correct time

$\underline{360 \times 2} = \underline{720 \text{ days}}$

Logical Reasoning

*

Name	Age	State
Pr	44	GJ
La	18	MH
AS	32	MP
Mi	38	Or
Pa	34	Ka
Ku	26	UP

*